



Fundamentals of Optics

Lecturer

YanJun LIU (刘言军)

E-mail: yjliu@sustech.edu.cn

Room 239, South Tower, College of Engineering

Department of Electrical and Electronic Engineering

Southern University of Science and Technology

Shenzhen, Guangdong, China

Teaching Assistant

Ye LI (李烨)

E-mail: 12250018@mail.sustech.edu.cn

Room 237, North Tower, College of Engineering

Ziyan LEI (雷子言)

E-mail: 12443025@mail.sustech.edu.cn

Room 338, South Tower, College of Engineering

Q&A time: 2-4 pm on Monday

Room 239, South Tower, College of Engineering

Rules

- Final exam: 30%
- Mid-term exam: 30%
- Homework: 25%
- Attendance: 15%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

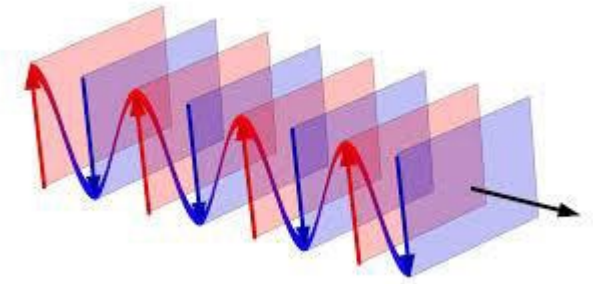
CHAPTER 2

ELECTROMAGNETIC THEORY, PHOTONS, AND LIGHT

Review

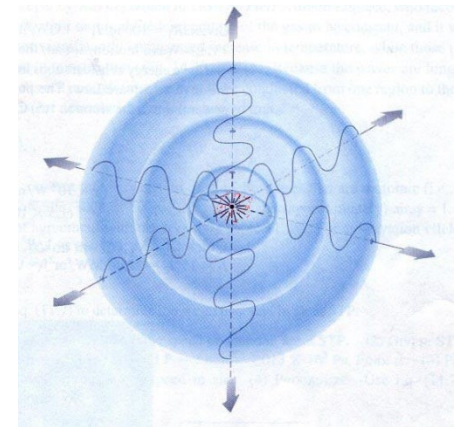
➤ Plane wave

$$\psi(\vec{r}) = Ae^{i(\vec{k} \cdot \vec{r} \mp \omega t)}$$



➤ Spherical wave

$$\psi(r, t) = \left(\frac{A}{r}\right) \cos(kr \mp \omega t)$$



➤ Basic parameters of a harmonic wave

Phase velocity $v = \frac{\omega}{k}$

Propagation vector $k = 2\pi/\lambda$

Period $\tau = \lambda/v$

➤ One-dimensional **differential wave equation**

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

➤ Three-dimensional **differential wave equation**

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

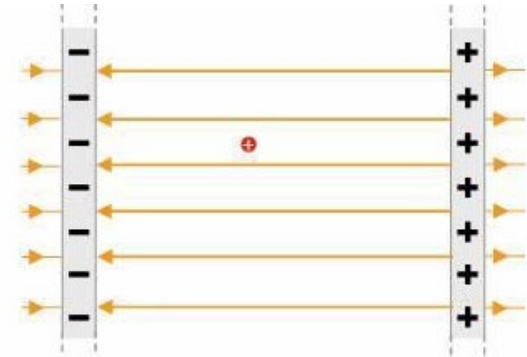
Wave-particle manifestations

- ❑ Classical electrodynamics unalterably leads to the picture of a continuous transfer of energy by ways of electromagnetic waves.
- ❑ Quantum electrodynamics describes electromagnetic interactions and the transport of energy in terms of massless elementary “particles” known as photons.
- ❑ Geometrical optics: the wavelength of light is small in comparison to the size of the apparatus (lenses, mirrors, etc.)

2.1 Basic laws of electromagnetic theory

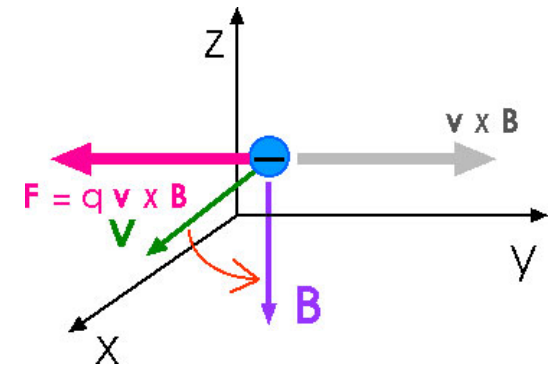
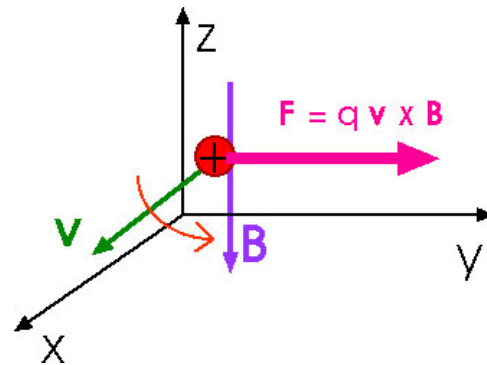
□ Electric field $\vec{E}$

$$\vec{F}_E = q \cdot \vec{E}$$



□ Magnetic induction or magnetic field $\vec{B}$

$$\vec{F}_M = q \cdot \vec{v} \times \vec{B}$$



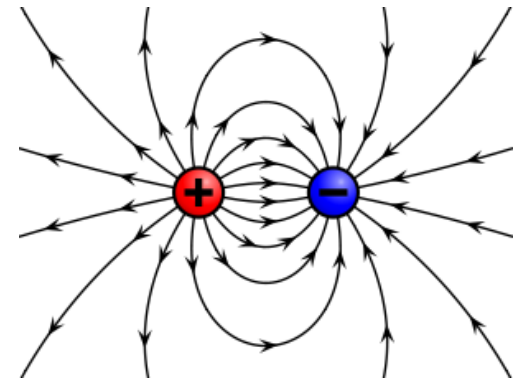
Left-hand rule

Source

❑ Electric fields are generated by

Electric charges

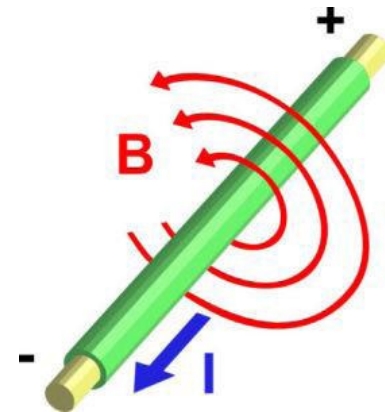
Time-varying magnetic fields



❑ Magnetic fields are generated by

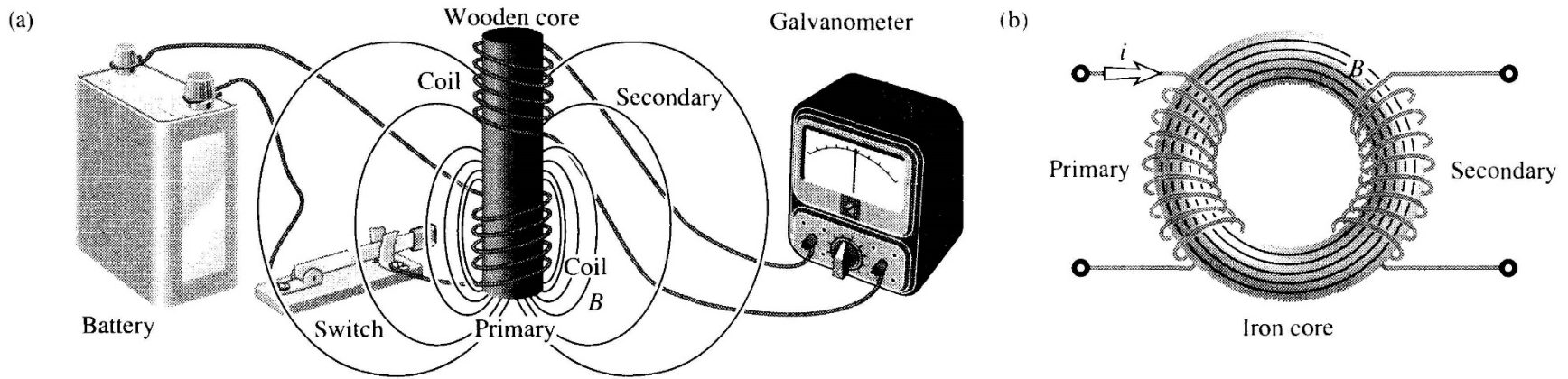
Electric currents

Time-varying electric fields



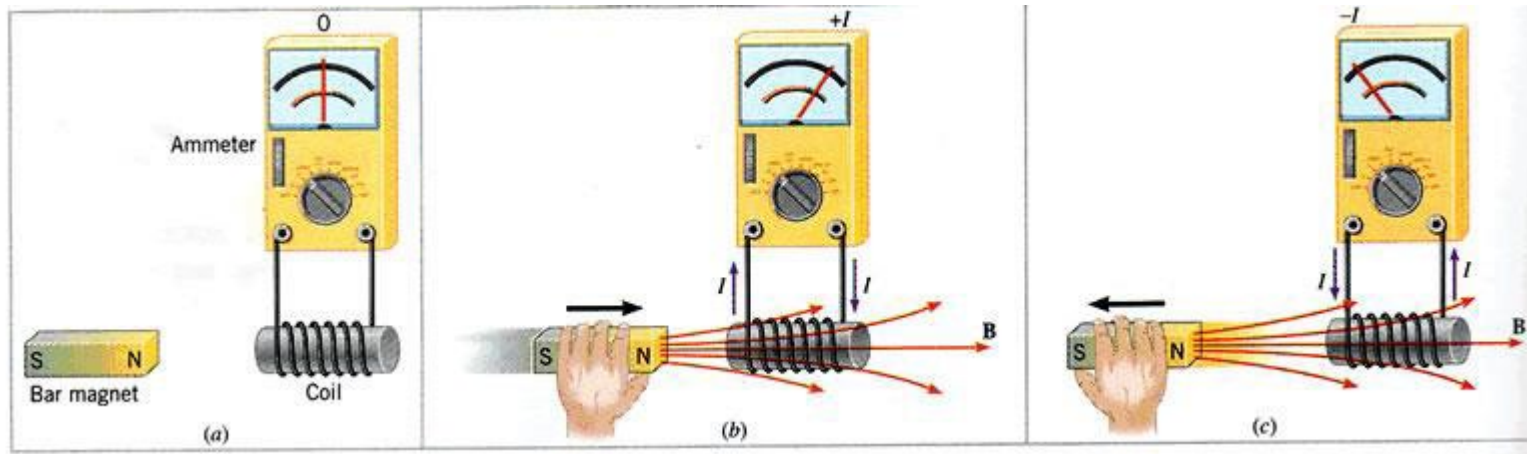
2.1.1 Faraday's Induction Law

□ A changing magnetic field generated a current



(a) The start of a current in one coil produces a time varying magnetic field that induces a current in the other coil. (b) An iron core couple the primary coil to the secondary

□ By thrusting a magnet into a coil, Faraday showed that there is a voltage known as the induced electromotive force (emf, 感应电动势) across the coil's terminals.



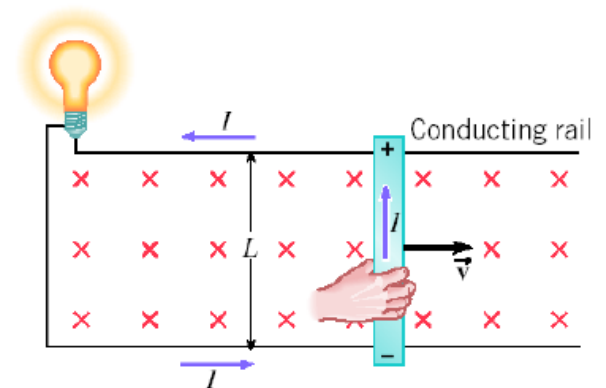
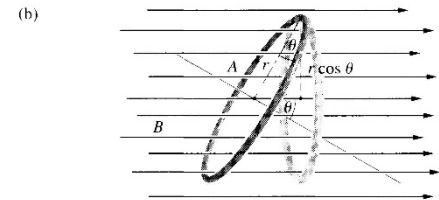
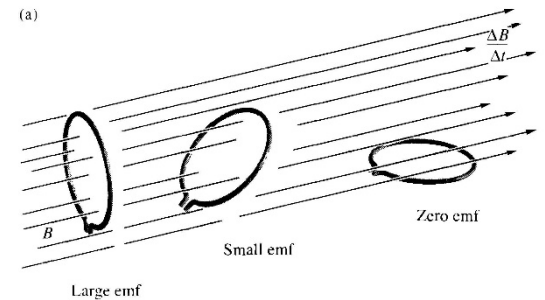
The induced emf depends on the rate-of-change of B through the coil and not on B itself.

- When the B is changing, the induced emf is proportional to the area A of the loop penetrated perpendicularly by the field.

$$emf \propto A_{\perp} \Delta B / \Delta t$$

- When the field is constant, the induced emf is proportional to the rate-of-change of the perpendicular area penetrated

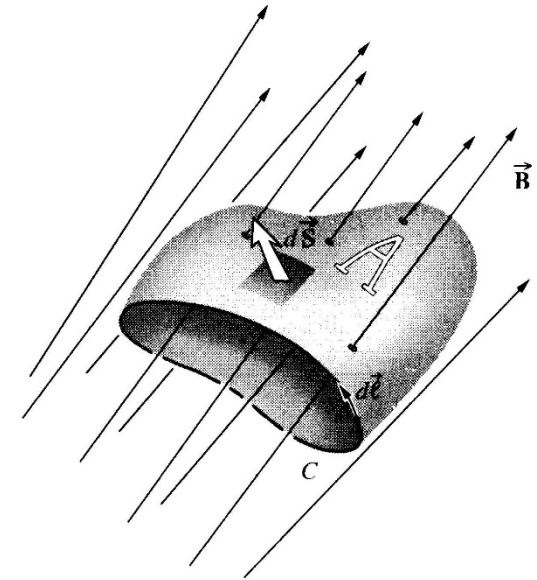
$$emf \propto B \Delta A_{\perp} / \Delta t$$



❑ The emf depends on the rate-of-change of both $A_{\perp}$ and B

❑ Flux of the magnetic field (Φ_M)

$$\Phi_M = B_{\perp} A = BA_{\perp} = BA \cos \theta$$



❑ More generally,

$$\Phi_M = \iint_A \vec{B} \cdot d\vec{S}$$

❑ The induced emf:

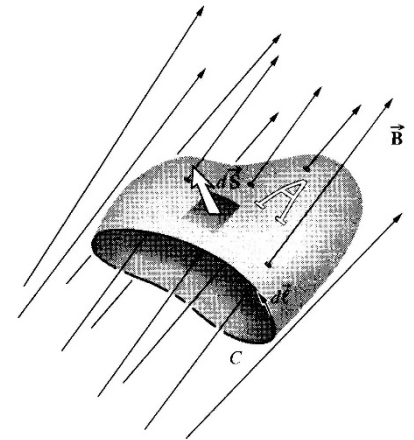
$$emf = -\frac{d\Phi_M}{dt}$$

- emf: a potential difference (potential-energy difference per unit charge; work done per unit charge; force per unit charge times distance, electric field times distance):

$$emf = \oint_C \vec{E} \cdot d\vec{\ell}$$

- Faraday's Induction Law

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$



- As to electromagnetic waves traveling in space where there are no wire loop, so the induction law:

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\iint_A \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

A time-varying magnetic field will have an electric field associated with it.

2.1.2 Gauss's Law- Electric

- ❑ The volume flux (Av) through both end surfaces is equal in magnitude.

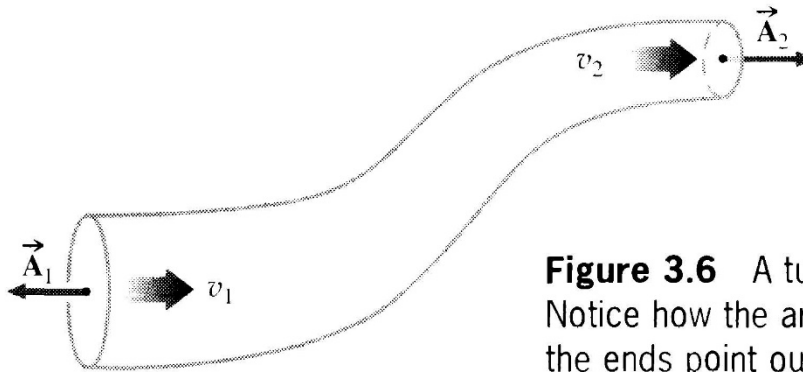
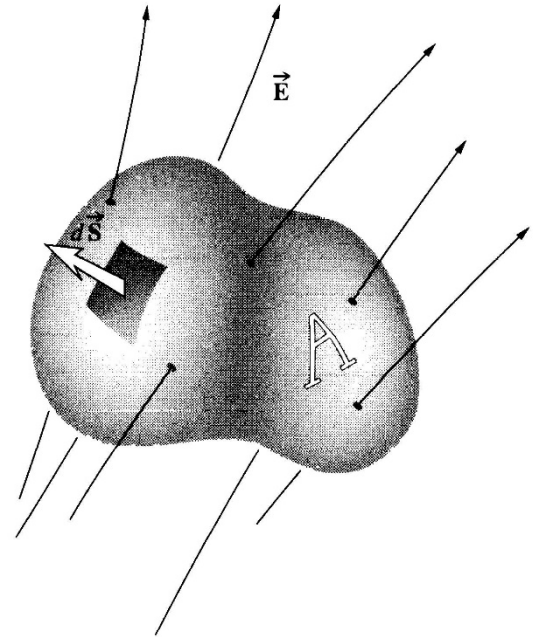


Figure 3.6 A tube of fluid flow. Notice how the area vectors on the ends point outward.

- ❑ When a small pipe is inserted into the region either sucking out fluid (a sink) or delivering fluid (a source), the net flux will be nonzero.

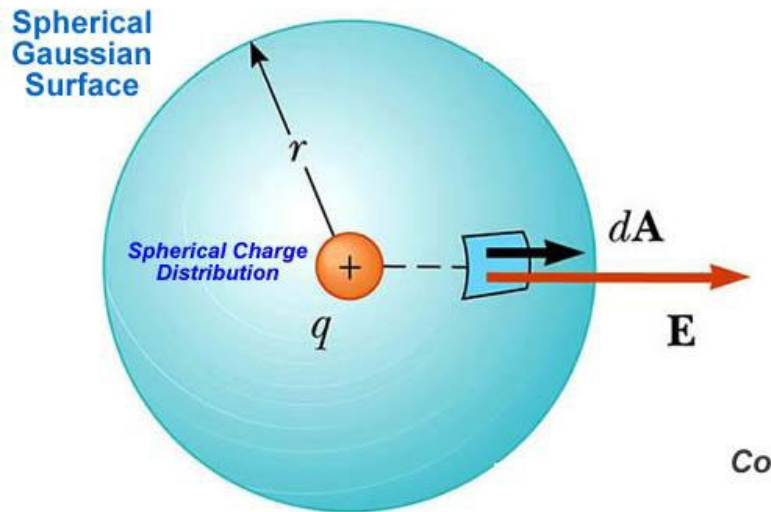
- The flux of the electric field through A:

$$\Phi_E = \oint_A \vec{E} \cdot d\vec{S}$$



- When there are no sources or sinks of the electric field within the region encompassed by the closed surface, the net flux through the surface equals zero.

- ❑ What happened in the presence of the internal sources and sinks:
- ❑ Consider spherical surface of radius r centered on a positive point-charge (q) in vacuum.



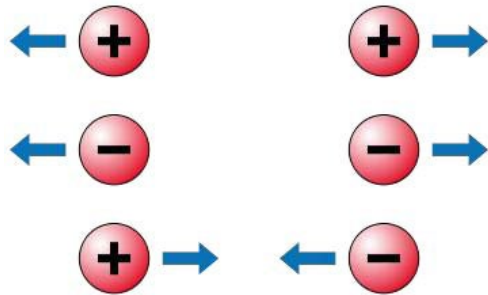
Gauss's Law

$$\oint_A \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \iiint_V \rho dV$$

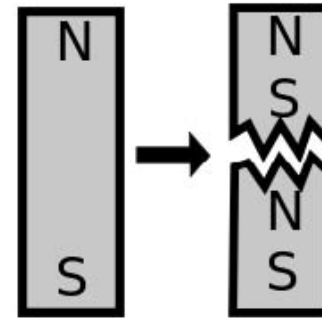
ϵ_0 : electric permittivity (电容率) of free space and equals to $8.8542 \times 10^{-12} \text{C}^2/(\text{N} \cdot \text{m}^2)$

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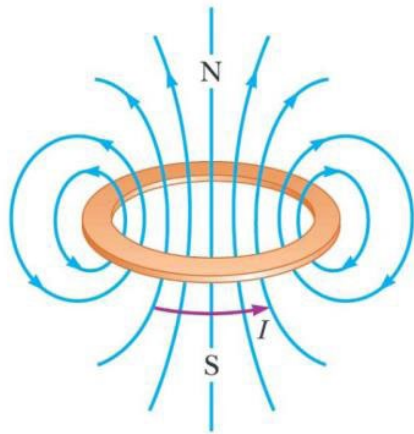
2.1.3 Gauss's Law- Magnetic



Electric charge



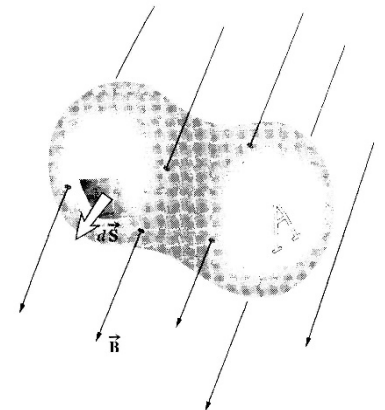
Magnetic poles



Magnetic fields can be described in terms of current distributions

Magnetic equivalent of Gauss's Law

$$\Phi_M = \oiint_A \vec{B} \cdot d\vec{S} = 0$$



2.1.4 Ampere's Circuital Law

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- Imagine a straight current-carrying wire in the vacuum and the circular B-field surrounding it.
- From experiments

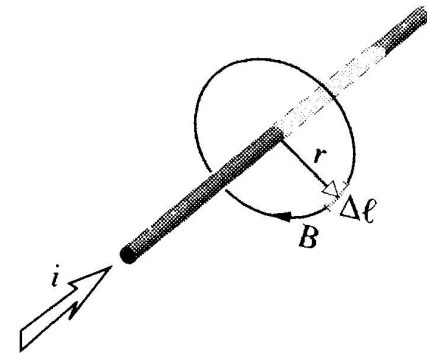
$$B = \mu_0 i / 2\pi r$$

Ampere's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \sum i$$

Ampere's Law in terms of the current density

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \vec{J} \cdot d\vec{S}$$



The $\vec{B}$ -field surrounding a current-carrying wire.

Ampere's Law: is it the truth?

❑ Consider a capacitor charging:

As to area A_1 : the above
Ampere's law is truth.

As to area A_2 : it is not truth

When charging or discharging a capacitor, one can measure a B -field around the wire and in the region between its plates

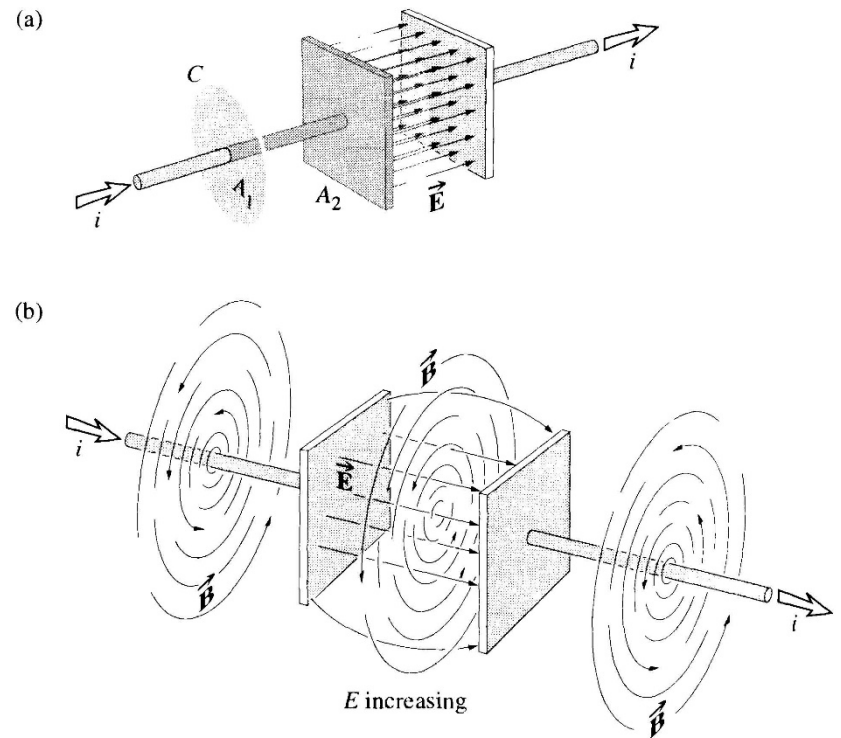
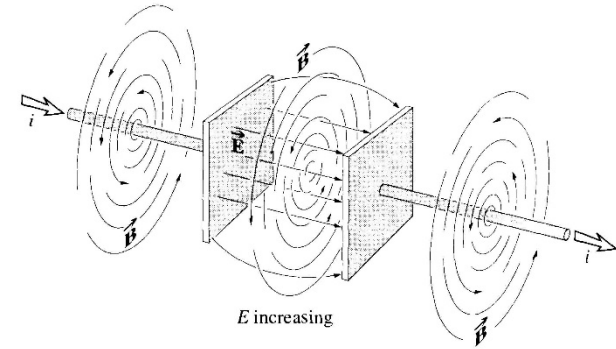


Figure 3.11 (a) Ampère's Law is indifferent to which area A_1 or A_2 is bounded by the path C . Yet a current passes through A_1 and not through A_2 , and that means something is very wrong. (b) $\vec{B}$ -field concomitant with a time-varying $\vec{E}$ -field in the gap of a capacitor.

□ The electric field of a capacitor

$$E = \frac{Q}{\epsilon A}$$

A: the area of each plate;
Q: the charge on it.



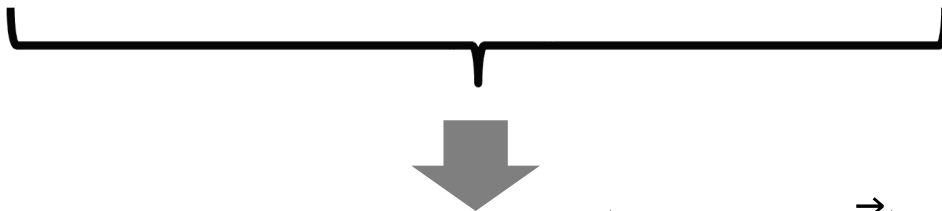
□ As the charge varies, the electric field changes

$$\epsilon \frac{\partial E}{\partial t} = \frac{i}{A}$$

□ Maxwell's contribution: define a displacement current density

$$\vec{J}_D \equiv \epsilon \frac{\partial \vec{E}}{\partial t}$$

- ❑ Combine Ampere's law and Maxwell's displacement current density:

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \vec{J} \cdot d\vec{S} \quad \vec{J}_D \equiv \varepsilon \frac{\partial \vec{E}}{\partial t}$$


$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \left(\vec{J} + \varepsilon \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$

μ_0 : the permeability (磁导率) of free space and it is defined as $4\pi \times 10^{-7} \text{ N} \cdot \text{s}^2 / \text{C}^2$

- ❑ When $\vec{J} = 0$, a time-varying $\vec{E}$ -field will be accompanied by a $\vec{B}$ -field.

2.1.5 Maxwell's equations

□ Faraday's Induction Law

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

□ Gauss's Law- Electric

$$\oiint_A \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \iiint_V \rho dV$$

□ Gauss's Law- Magnetic

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

□ Ampere-Maxwell's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \left(\vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$

□ Maxwell's equations in free space

$$\epsilon = \epsilon_0; \mu = \mu_0; \text{ and } \rho = 0; J = 0$$

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\iint_A \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

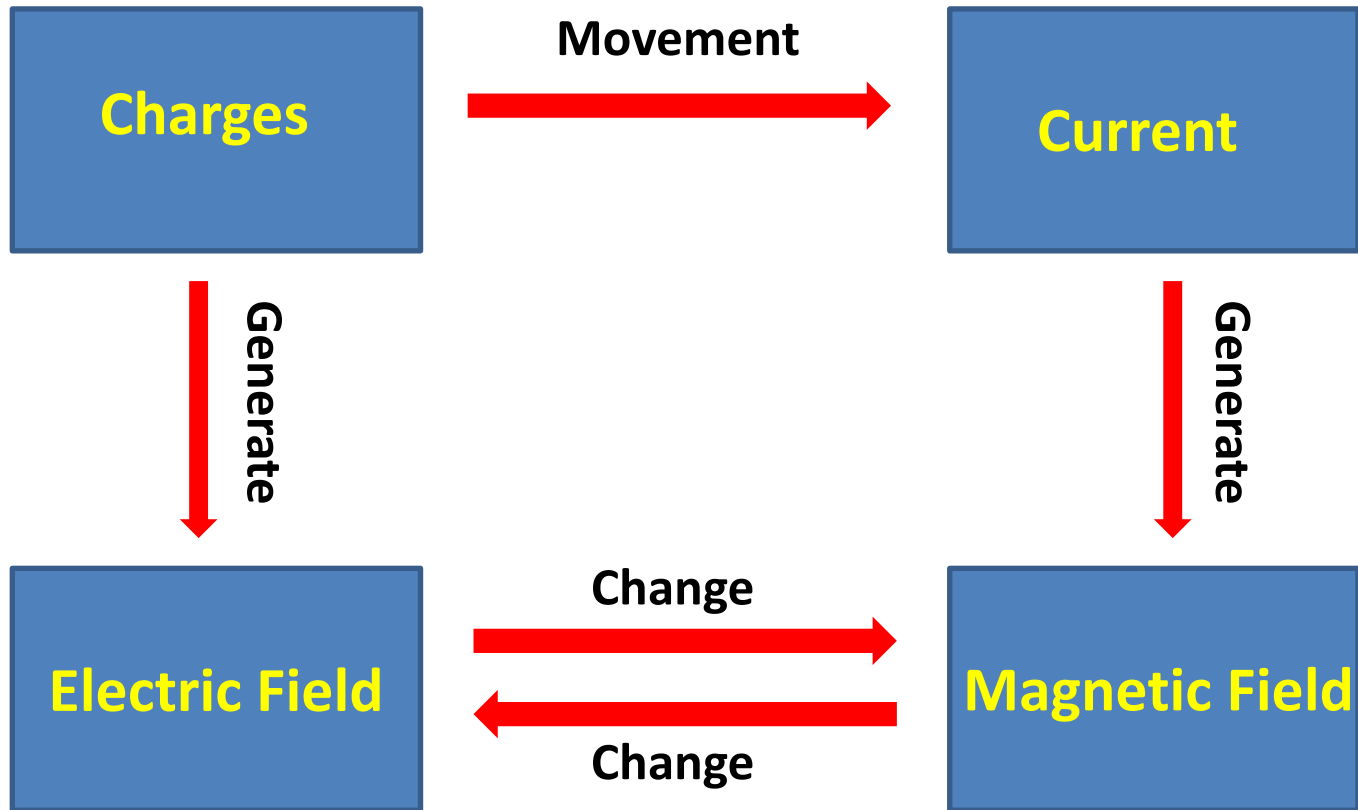
$$\oiint_A \vec{E} \cdot d\vec{S} = 0$$

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \epsilon_0 \iint_A \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S}$$

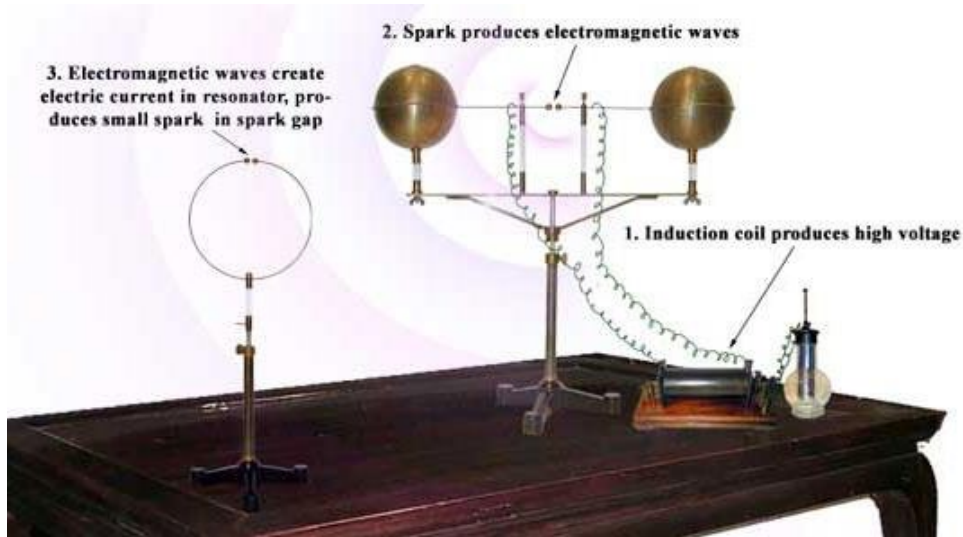
□ A time-varying magnetic field generates an electric field and a time-varying electric field generates a magnetic field.

Link between electric field and magnetic field



Hertz's Experiment

Video link



- ❑ Hertz said “It's of no use whatsoever, this is just an experiment that proves Maestro Maxwell was right.”
- ❑ Marconi and Popov did their greatest contribution ([wireless telegraph](#)) to the application of electromagnetic wave.